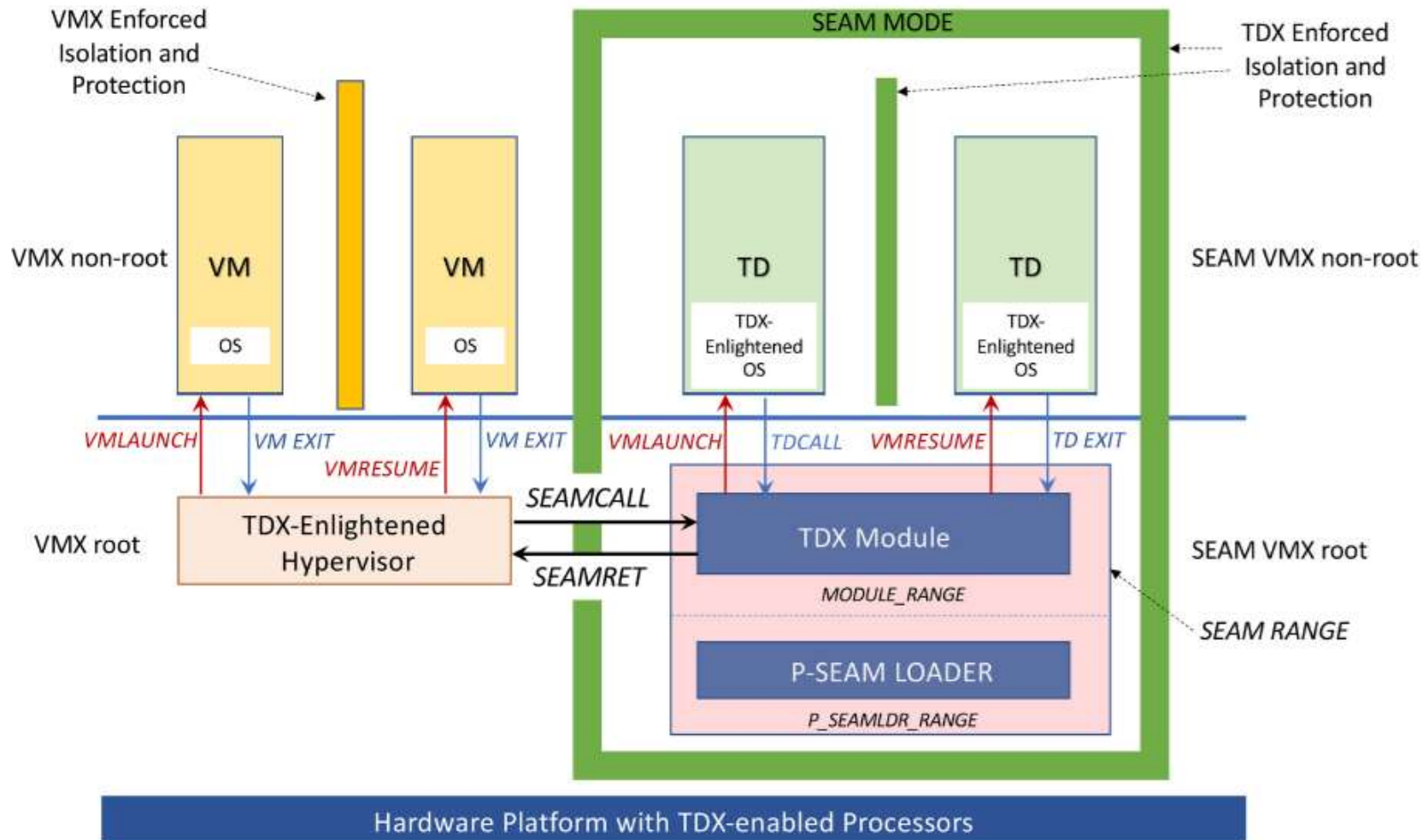


TDX System Architecture



- SEAM VMX Non-Root Mode
 - Trust Domains (TDs) execute in this mode with encrypted CPU state and memory
- SEAM VMX Root Mode
 - TDX Module executes in this mode, handling SEAMCALL instructions from hypervisor
- Mode Transitions
 - SEAMCALL/SEAMRET instructions enable secure transitions between modes

Memory Partitioning

■ Memory Partition

- Normal memory
- Secure memory for storing sensitive data
- Shared memory for I/O (not protected)

■ HKID Space Partitioning

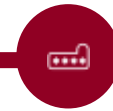
- The HKID space is partitioned at boot into private HKIDs (for TDs and TDX Module) and shared HKIDs (for legacy VMs and host)
- Each TD receives a unique private HKID bound to an ephemeral encryption key.

Memory Confidentiality and Integrity



Logical Integrity (Li)

Uses **TD Owner Bit** to detect unauthorized writes at software level.
Bit is set to 1 for TD writes, cleared to 0 for external writes.



Cryptographic Integrity (Ci)

Computes **28-bit MAC** on each cache line write using 128-bit MAC key. Recalculates MAC on read to detect tampering.

Both modes mark tampered memory as "poisoned." Reading poisoned content triggers TD exit (for TD) or disables TDX Module (for Module reads). However, neither can detect memory replay attacks if both content and metadata are rolled back.

TDX I/O Model

■ Security Requirements

- Hypervisors and peripheral devices are untrusted and cannot access TD private memory.
- TDs must explicitly share memory for I/O by setting the shared bit in Guest Physical Address (GPA).

■ TD Responsibilities

- Seal I/O data before leaving trust boundary
- Place sealed data in shared memory
- Validate all incoming I/O data
- Use hardened drivers only

Remote Attestation

